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Hilbert Geometry

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To the Instructor

The non-Euclidean revolution of the nineteenth century culminated in the appearance of David Hilbert's 1899 *Grundlagen der Geometrie*, at last resolving the problems with Euclid's system that mathematicians had struggled with for centuries. I wrote these notes for a sophmore-junior level course in axiomatic geometry based on a version of Hilbert's axioms. They are intended for students who already have a working knowledge of Euclidean geometry and would like a deeper understanding of its foundations. Students in my classes come with a year of calculus and at least one foundational theorem proving course. Like any course based on Hilbert's axioms, these notes offer a rigorous axiomatic development at the cost of not covering more advanced topics in geometry per se. For example, these notes are confined to neutral geometry in which no parallel axiom is assumed.

Chapter 1 presents diverse models of geometry that play a central role throughout the course. I emphasize the fact that a statement of geometry is only true or false in a model; that as we add axioms, we eliminate models, thereby narrowing the discourse to those models which reflect what most interests us. Several times I tried to introduce the projective plane as pairs of antipodal points of the sphere, but found this too conceptually difficult for my audience and decided to replace it with the sphere model in which betweenness is not fully defined.

As we go into Chapter 2 students quickly we see how models can guide their proofs. For example, Pasch's Theorem is clearly false in $\mathbb{R}^3$; therefore any proof of this theorem would necessarily utilize some axiom that fails in $\mathbb{R}^3$. Along the way I present the bumpy planes which are betweenness isomorphic to $\mathbb{R}^2$ and satisfy all of the congruence axioms except SAS. This illustrates that fact that SAS is really saying that the curvature of the space is uniform, which significantly limits the possible models. Here I like to mention the Hyperbolic Plane, positive and negative curvature, and how they relate to the parallel axioms, to recent developments in cosmology, and to the structure of the universe. I do this judiciously, when no one has something to present and interest seems to be sufficient. Without going deeply into the content of geometry, I find that, by the end of Chapter 4, students have a solid working sense of how geometry is developed. With this infrastructure in place, one could easily add further theorems for a class

that has time to handle more material.

My use of inquiry-based learning generally follows four steps.

1. I present the necessary concepts and definitions, and then state and motivate the problems and theorems in the text. This is an interactive process in which students attempt to translate intuitive concepts, illustrated by pictures, into precise definitions. They then take those precise definitions and guess what axioms and theorems are true about them.

2. Students work outside class to solve the problems and prove the theorems that were presented. They need to write these solutions and proofs out in a form that will allow them to explain them to the class.

3. Most class time is spent with individual students presenting their solutions and proofs at the board. As appropriate, presentations are followed by discussion until either everyone agrees it is correct or errors are found that need to be corrected for a subsequent presentation. Errors students make invariably provide a valuable learning experience for the entire class. Students learn that errors are part of the discovery process, and simply mean that the process is not yet complete.

4. Once a satisfactory solution or proof is presented, all students have the task of writing up a final draft of it that they will add to a personal portfolio. I review the portfolios on a regular basis to ensure that students are all keeping up. The course will end with each student having a completed portfolio consisting of a solution of each problem and a proof of each theorem that appears in the text.

Fourteen axioms are presented as they are needed in these notes, and are listed in the order they arise in an appendix. The first and second 4-Point Axioms are actually redundant, as they can be proven once the the Plane Separation Axiom is available.

These notes evolved over many iterations, and now work well for a one semester course at SUNY New Paltz. Depending on your audience, you may either need to include additional material and remove unnecessary intermediate lemmas, or delete material and add further intermediate lemmas.

David Clark
August, 2011

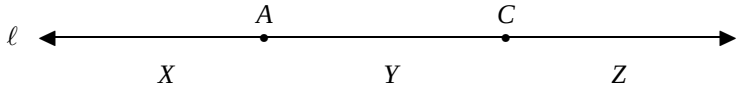
Chapter 1

Models of Geometry

A ***model*** $\mathbb{G}$ ***of geometry*** consists of a set of objects which we call ***points*** and a ***betweenness relation*** among triples of points. To specify a model $\mathbb{G}$ we must therefore

- specify the set of points of $\mathbb{G}$ and
- tell which points are between which other points.

If A, B and C are points, we will write "$[ABC]$" as an abbreviation for the statement that B is between A and C. Given two points A and C, the ***line*** $\ell = \overleftrightarrow{AC}$ is the set of points consisting of A, C, all points X such that $[XAC]$, all points Y such that $[AYC]$, and all points Z such that $[ACZ]$.



Here are some examples. For each model we define *point* and *between* and then describe the lines.

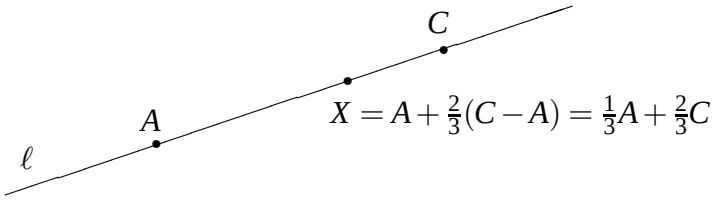
$\mathbb{R}^2$ (Euclidean 2-Space)

Our first example is the model that is most familiar. A *point* is an ordered pair (x,y) of real numbers. Points of $\mathbb{R}^2$ can be added and multiplied by real numbers using the definitions

$$(a,b)+(c,d) = (a+c,b+d)$$
$$t(a,b) = (ta,tb)$$

If A and C are distinct points of $\mathbb{R}^2$, then a point B is *between* A and C if there is a number t between 0 and 1 such that

$$B = A+t(C-A) = (1-t)A+tC$$



The line $\ell = \overleftrightarrow{AC}$ will then consist of the set of points

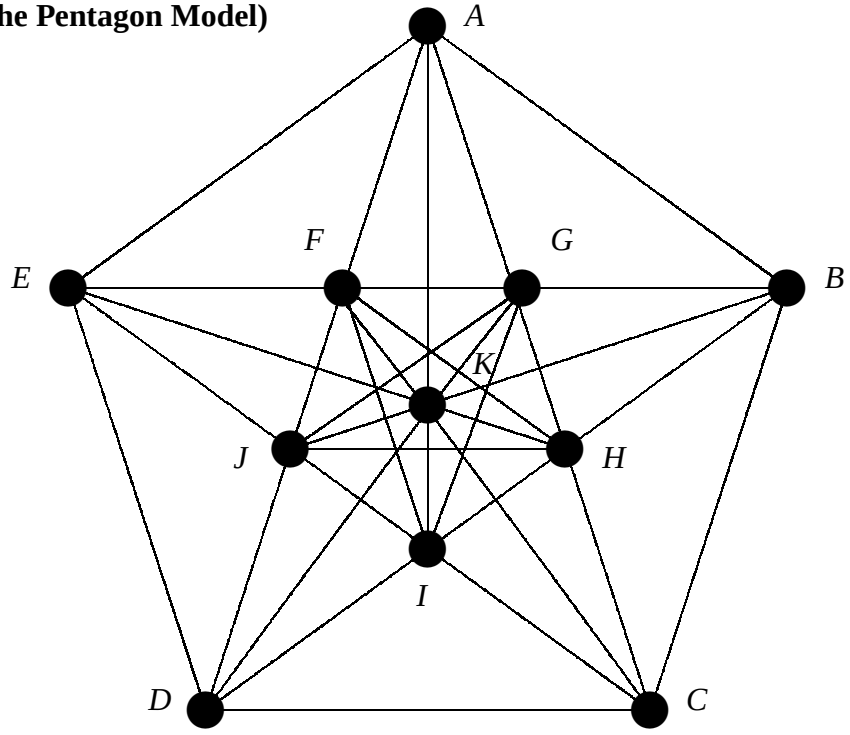
$$\ell := \{X \mid X = A + t(C-A) \text{ for some } t \in \mathbb{R}\},$$

where t ranges over *all* values in the set $\mathbb{R}$ of real numbers.

$\mathbb{D}$ (the Disc Model)

A *point* of $\mathbb{D}$ is a point (x,y) of $\mathbb{R}^2$ where $x^2+y^2<1$. Given points A, B and C of $\mathbb{D}$, we say that B is *between* A and C if $[ABC]$ is true in $\mathbb{R}^2$. A line of $\mathbb{D}$ will then consist of the intersection of a line of $\mathbb{R}^2$ with the points of $\mathbb{D}$, provided this intersection contains at least two points.

$\mathbb{P}$ (the Pentagon Model)



This model has only 11 *points* which are labeled A through K in the diagram. A point Y is *between* points X and Z if X, Y and Z are distinct points and Y appears between X and Z. Model $\mathbb{P}$ has exactly 20 lines, each with either

2, 3 or 4 points. For example, $\{E,A\}$ and $\{E,F,G,B\}$ are lines of $\mathbb{P}$ with $[BFE]$.

$\mathbb{Q}^2$ (the Rational Plane)

A *point* of $\mathbb{Q}^2$ is a point of $\mathbb{R}^2$ whose coordinates are both rational numbers. For rational points A, B and C, we say that B is *between* A and C if $[ABC]$ is true in $\mathbb{R}^2$.

$\mathbb{R}^3$ (Euclidean 3-Space)

As a model of geometry, three dimensional Euclidean space is very similar to two dimensional Euclidean space. A *point* of $\mathbb{R}^3$ is an ordered triple (x,y,z) of real numbers. Points of $\mathbb{R}^3$ can be added and multiplied by real numbers using the definitions

$$(a,b,c)+(d,e,f)=(a+d,b+e,c+f)$$
$$t(a,b,c)=(ta,tb,tc)$$

If A and C are distinct points of $\mathbb{R}^3$, then a point B is *between* A and C if there is a number t between 0 and 1 such that

$$B=A+t(C-A)=(1-t)A+tC$$

The line $\ell=\overleftrightarrow{AC}$ will then consist of the set of points

$$\ell:=\{X \mid X=A+t(C-A) \text{ for some } t\in\mathbb{R}\},$$

where t ranges over *all* values in the set $\mathbb{R}$ of real numbers.

$\mathbb{S}$ (the Sphere)

Consider the sphere $\mathbb{S}$ in $\mathbb{R}^3$ with center $(0,0,0)$ and radius 1. A *point* of $\mathbb{S}$ is a point (x,y,z) of $\mathbb{R}^3$ satisfying the equation $x^2+y^2+z^2=1$. A ***great circle*** of the sphere is a circle with center $(0,0,0)$ and radius 1. Point B is *between* points A and C if A and C are not opposite points ($A\neq -C$), and B lies along the shorter great circle arc from A to C.

The lines of $\mathbb{S}$ are then exactly the great circles. Notice that every line of $\mathbb{S}$ contains distinct points A, B and C no one of which is between the other two.

Later, we will include additional models of geometry. Notice that the terms *point* and *between* do not have any fixed meaning in the context of geometry. Rather, they are only defined in a particular model of geometry. For this reason they are often referred to, somewhat misleadingly, as ***undefined terms***. In contrast, ***defined terms*** are terms like ***line*** that are defined

generally, for all models at once, using the undefined terms. Here are a few other examples of defined terms.

We say that s is a ***segment*** if there are two points A and B such that

$$s = \{X \mid X = A \text{ or } [AXB] \text{ or } X = B\}.$$

We use the notation $s = AB$. We say that $\overrightarrow{r}$ is a ***ray*** if there are two points A and B such that

$$\overrightarrow{r} = \{X \mid X = A \text{ or } [AXB] \text{ or } X = B \text{ or } [ABX]\}.$$

We use the notation $\overrightarrow{r} = \overrightarrow{AB} = \overleftarrow{BA}$. For completeness, we say (again) that ℓ is a ***line*** if there are two points A and B such that

$$\ell = \{X \mid [XAB] \text{ or } X = A \text{ or } [AXB] \text{ or } X = B \text{ or } [ABX]\}.$$

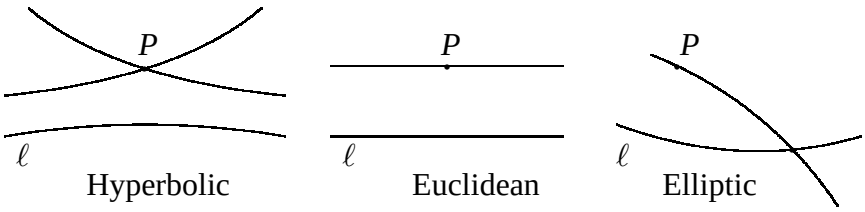
We use the notation $\ell = \overleftrightarrow{AB}$.

Lines ℓ and m are said to be ***parallel*** if they have no point in common; that is, $\ell \cap m = \varnothing$. A set of points is ***collinear*** if some line contains them all. The issue of the existence and uniqueness of parallel lines has played a central role in the history of geometry, and will be a theme we will follow in this course. It turns out that in most interesting models of geometry, parallel lines exist in one of the following three regular patterns. Notice that a model of geometry can have at most one of these properties.

Hyperbolic Parallel Property: For every line ℓ and point $P \notin \ell$, there are at least two lines containing P that are parallel to ℓ.

Euclidean Parallel Property: For every line ℓ and point $P \notin \ell$, there is exactly one line containing P that is parallel to ℓ.

Elliptic Parallel Property: For every line ℓ and point $P \notin \ell$, there is no line containing P that is parallel to ℓ; that is, there are no parallel lines at all.



We will be interested in knowing which, if any, of these parallel properties hold in each of the geometries that we will study.

In the following problems, state and justify each answer. Either prove or disprove each statement.

Problem 1. $\mathbb{R}^2$ ***(Euclidean** 2**-Space)***

(i) *The point $B = (16,11)$ is between $A = (5,7)$ and $C = (49,23)$.*

(ii) *The point $B = (15,12)$ is between $A = (4,8)$ and $C = (50,22)$.*

Problem 2. $\mathbb{D}$(***the Disc Model***)

(i) *Does a line of $\mathbb{R}^2$ that contains a point of $\mathbb{D}$ necessarily contain at least two points of $\mathbb{D}$?*

(ii) *How many lines are there containing the point $P = (0,0.5)$ that are parallel to the line $\ell = \{(x,0) \mid -1 < x < 1\}$?*

Problem 3. $\mathbb{P}$ (***the Pentagon Model***)

(i) *List all lines that are parallel to the line $\ell = \{E,F,G,B\}$.*

(ii) *How many triples (X,Y,Z) of points are there where $[XYZ]$?*

(iii) *List all of the points of the rays $\overrightarrow{CH}$ and $\overrightarrow{CG}$.*

Problem 4. $\mathbb{Q}^2$ (***the Rational Plane***) *Since $\mathbb{Q}^2$ is contained in $\mathbb{R}^2$, two points A and B of $\mathbb{Q}^2$ define both line $\overleftrightarrow{AB}^Q$ of $\mathbb{Q}^2$ and line $\overleftrightarrow{AB}^R$ of $\mathbb{R}^2$.*

(i) *Show that $\overleftrightarrow{AB}^Q = \{X \mid X = A + t(B-A)$ such that $t \in \mathbb{Q}\}$, where $\mathbb{Q}$ is the set of rational numbers.*

(ii) *Show that $\overleftrightarrow{AB}^Q = \overleftrightarrow{AB}^R \cap \mathbb{Q}^2$.*

(iii) *Let $\overleftrightarrow{AB}^Q$ and $\overleftrightarrow{CD}^Q$ be lines in $\mathbb{Q}^2$. Is it true that $\overleftrightarrow{AB}^Q$ and $\overleftrightarrow{CD}^Q$ are parallel in $\mathbb{Q}^2$ if and only if $\overleftrightarrow{AB}^R$ and $\overleftrightarrow{CD}^R$ are parallel in $\mathbb{R}^2$?*

Problem 5. $\mathbb{R}^3$ (***Euclidean** 3**-Space***) *The model $\mathbb{R}^2$ has the property that two lines parallel to the same line are parallel to each other. Find a specific counterexample to show that this is not true in $\mathbb{R}^3$.*

Problem 6. $\mathbb{S}^2$ (***The Sphere***) *Consider the 3 collinear points $A = (1,0,0)$, $B = (0,1,0)$, and $C = (0,-1,0)$, lying on the line $\ell = \{(x,y,0) \mid x^2 + y^2 = 1\}$. Make 4 sketches of ℓ to illustrate segments AB and AC and rays $\overrightarrow{AB}$ and $\overrightarrow{AC}$. (Since ℓ lies entirely in the xy-plane, that is all you need to draw.)*

Problem 7. *Which of our models of geometry satisfy the Hyperbolic Parallel Property? the Euclidean Parallel Property? the Elliptic Parallel Property? Which satisfy none of these properties?*

There are many models of geometry all around us. If we start with any surface, we can take individual locations on the surface as our *points*, and say that point B is *between* A and C if the shortest path on the surface from A to C passes through B.

We will be interested in discovering the properties of these models. Our first inclination is to do this by studying models one by one, but this approach is extremely inefficient. Instead, we will adopt what is called the ***axiomatic method***. The idea is simple. We will first identify a small list of properties that are common to the models we are interested in studying. These properties will be called ***axioms***. We will then focus on discovering new properties that follow logically from the axioms. These will be called ***propositions*** and ***theorems***.

The economy of this method lies in the fact that, once we check that our axioms hold in a particular model, we are assured that every proposition or theorem that we prove from those axioms also holds in that model. The beauty of this method lies in the discovery that a few simple axioms can spawn a multitude of deep and complex propositions and theorems. To realize this discovery, you will need to use the definitions and axioms we give to prove the propositions and theorems in the order they are presented.

Chapter 2

Linear Geometry

In this chapter we will study the properties of points lying on a single line, and the betweenness relation among them. Points are ***collinear*** if they are all contained in one line.

Proposition 8. *If $[ABC]$, then A, B and C are collinear.*

Our first two axioms are about betweenness. We will add new axioms only as they are needed. (See the appendix for a complete list of the axioms.)

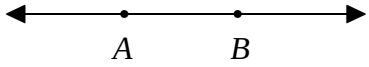
Symmetry Axiom. *If $[ABC]$, then A, B and C are distinct points and $[CBA]$.*

Uniqueness Axiom. *Given three points, at most one is between the other two.*

Proposition 9. *If A and B are two points, then $AB = BA$ and $\overleftrightarrow{AB} = \overleftrightarrow{BA}$.*

Proposition 10. *If A and B are two points, then*

1. $\overleftarrow{AB} \cup \overrightarrow{AB} = \overleftrightarrow{AB}$
2. $\overleftarrow{AB} \cap \overrightarrow{AB} = AB$



Collinearity Axiom. *If three points are collinear, then one of them is between the other two.*

Combining the last two axioms, we see that, for any three collinear points, exactly one is between the other two. For a finite set $A_1, A_2, \cdots, A_n$ of points, we will write $[A_1A_2\cdots A_n]$ as an abbreviation for the conjunction of all statements $[A_iA_jA_k]$ where $i < j < k$. Thus $[A_1A_2\cdots A_n]$ asserts that these

are n distinct collinear points which lie in the order that they are written. Our next two axioms tell how four points are arranged on a line.

1st 4-Point Axiom. *If $[ABC]$ and $[ACD]$, then $[ABCD]$.*

2nd 4-Point Axiom. *If $[ABC]$ and $[BCD]$, then $[ABCD]$.*

The next proposition is similar to Proposition 10, but requires our additional axioms.

Proposition 11. *If $[ABC]$, then*

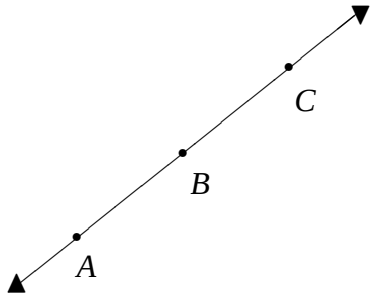
(i) $AB \cup BC = AC$

(ii) $AB \cap BC = \{B\}$

(iii) $\overleftarrow{AB} \cup \overrightarrow{BC} = \overleftrightarrow{AC}$

(iv) $\overleftarrow{AB} \cap \overrightarrow{BC} = \{B\}$

(v) $\overrightarrow{AB} = \overrightarrow{AC}$



Knowing how four points can be arranged on a line will allow us to prove that any two points on a line will determine that line. The proof of Proposition 12 requires many cases depending on how the points in question are arranged. Once you are convinced that the proofs of all the cases are very similar, you will probably not need to write them all down.

Proposition 12. *If $C, D \in \overleftrightarrow{AB}$ and $C \neq D$, then $\overleftrightarrow{CD} = \overleftrightarrow{AB}$.*

Theorem 13. *If A and B are two points, then $\overleftrightarrow{AB}$ is the only line containing them both.*

Infinity Axiom. *If B and D are distinct points, then there are distinct points A, C, and E on line $\overleftrightarrow{BD}$ such that $[ABCDE]$.*

Problem 14. *Show that every line contains at least 7 points, explaining how you know they are distinct.*

Problem 15. *Why do you think the last axiom called the "Infinity Axiom"?*

Problem 16. *Which of the models in Chapter 1 satisfy all of our present axioms, and which do not?*

Chapter 3

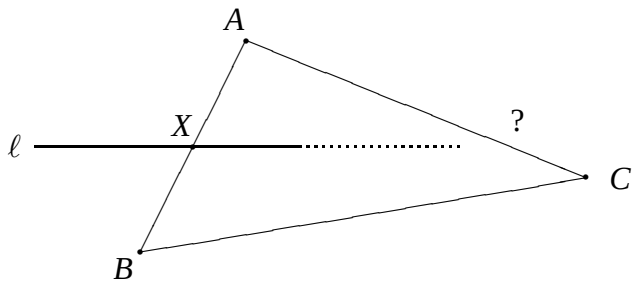
Plane Geometry

A ***triangle*** is the union of three segments AB, AC, and BC, where A, B, and C are non-collinear points. We denote this triangle by $\triangle ABC$. The points A, B, and C are called its ***vertices***, and the three segments are called its ***sides***.

We begin this chapter with an axiom that is equivalent to the assertion that there is more than one line.

Triangle Axiom. *There exists a triangle.*

Pasch's Theorem 17. *Assume that line ℓ intersects side AB of $\triangle ABC$ at a point X between A and B. Then ℓ also intersects either AC or BC. If C does not lie on ℓ, then ℓ does not intersect both AC and BC.*



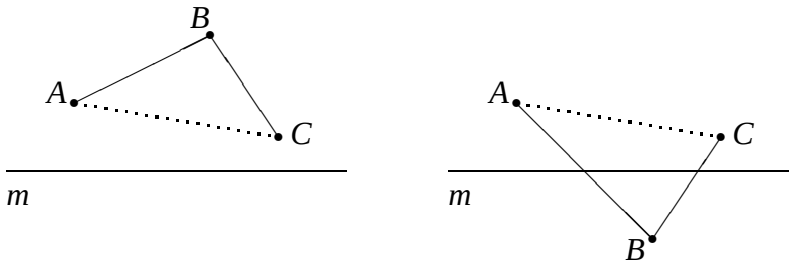
If you have difficulty proving this theorem, look back at the six models of Chapter 1. Is there one of them in which our axioms all hold, but Pasch's Theorem fails? If so, then we would know immediately that we couldn't possibly prove Pasch's Theorem from the axioms!

Our last betweenness axiom tells us that a model is exactly 2-dimensional. If m is a line and A and B are points not on m, we say that A and B are ***on the same side of*** m, written $A, B \mid_m$, if there is no point C on m such that $[ACB]$. When there is a point C on m such that $[ACB]$, we write $A \mid_m B$, and say that A and B are ***on opposite sides of*** m.

Plane Separation Axiom.

(i) *If $A,B\,|_m$ and $B,C\,|_m$, then $A,C\,|_m$.*

(ii) *If $A\,|_m\,B$ and $B\,|_m\,C$, then $A,C\,|_m$.*



This axiom says that every line separates the set of points into two half planes in the following sense.

Proposition 18. *If m is a line, then there are two points A and B not on m such that, for every point C not on m, either $A,C\,|_m$ or $B,C\,|_m$ but not both.*

Problem 19. *Determine which of the models $\mathbb{R}^2$, $\mathbb{D}$, $\mathbb{P}$, $\mathbb{Q}^2$, $\mathbb{R}^3$, and $\mathbb{S}^2$ satisfy the Plane Separation Axiom.*

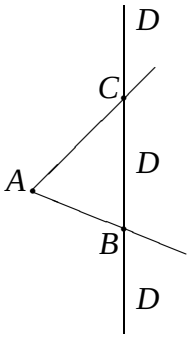
Here is a simple way to verify that two points are on the same side of a line.

Proposition 20. *Assume that $A \in \ell$, that $B,C \notin \ell$, and that $[A,B,C]$. Then $B,C\,|_\ell$.*

Now you should be able to prove Pasch's Theorem.

Rays $\overrightarrow{AB}$ and $\overrightarrow{CD}$ are ***opposite rays*** if A and C are the same point and A is between B and D. Let A, B, and C be non-collinear points. The ***angle*** $\angle CAB$ is the union of the non-opposite rays $\overrightarrow{AB}$ and $\overrightarrow{AC}$. The ***interior*** of $\angle CAB$ is the set of points D that are on the same side of $\overleftrightarrow{AB}$ as C and the same side of $\overleftrightarrow{AC}$ as B. The ***exterior*** of $\angle CAB$ is set of point E that are neither on one of the rays of $\angle CAB$ nor in the interior of $\angle CAB$.

Lemma 21. *Let D be a point lying on line $\overleftrightarrow{BC}$. Then D is in the interior of $\angle CAB$ if and only if it is between B and C.*



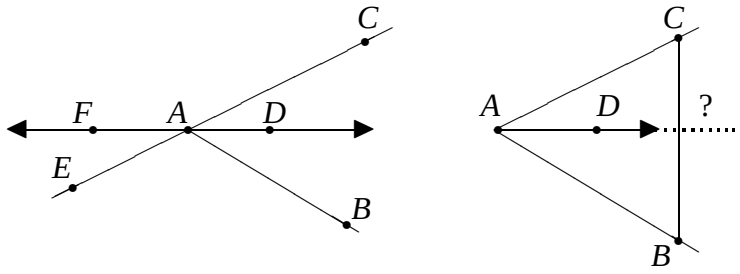
Lemma 22. *If D is in the interior of $\angle CAB$, with $[EAC]$ and $[FAD]$, then*

(i) *every point on ray $\overrightarrow{AD}$ other than A is in the interior of $\angle CAB$;*

(ii) *no point on the opposite ray $\overrightarrow{AF}$ is in the interior of $\angle CAB$;*

(iii) *B and C are on opposite sides of $\overleftrightarrow{AD}$.*

Ray $\overrightarrow{AD}$ is said to be ***between*** rays $\overrightarrow{AB}$ and $\overrightarrow{AC}$ if *A*, *B*, and *C* are not collinear, and *D* is in the interior of $\angle CAB$. (Notice that this definition would make no sense without part (i) of Lemma 22.)



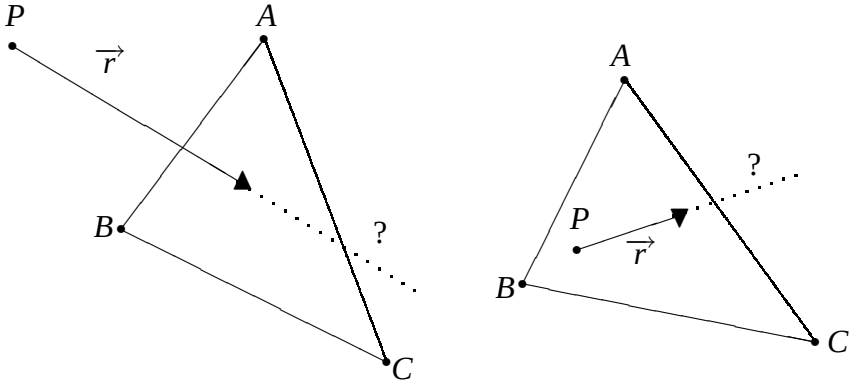
Crossbar Theorem 23. *If $\overrightarrow{AD}$ is between $\overrightarrow{AB}$ and $\overrightarrow{AC}$, then $\overrightarrow{AD}$ intersects the segment BC.*

The ***interior*** of $\triangle ABC$ is the set of points that are simultaneously on the *A* side of *BC*, the *B* side of *AC*, and the *C* side of *AB*; that is, in the intersection of the three angle interiors. The ***exterior*** of the triangle consists of all points neither on the triangle nor in its interior.

Proposition 24. *Let $\overrightarrow{r}$ be a ray emanating from a point P.*

(i) *If P is an exterior point of $\triangle ABC$ and $\overrightarrow{r}$ intersects side AB in a point between A and B, then $\overrightarrow{r}$ also intersects either side AC or side BC.*

(ii) *If P is an interior point of $\triangle ABC$, then $\overrightarrow{r}$ intersects one of the sides of $\triangle ABC$. If $\overrightarrow{r}$ does not pass through a vertex, then it intersects only one side.*



Chapter 4

Congruence

In this chapter, we study the notions of *congruence of segments* and *congruence of angles* in arbitrary models of geometry. These notions, along with *point* and *between*, must be specified in each model.

We will write

$$AB \cong CD$$

to mean that the segments AB and CD are congruent. In each of our models $\mathbb{R}^2$, $\mathbb{R}^3$, $\mathbb{Q}^2$, $\mathbb{D}$, and (locally) $\mathbb{S}$, we define segments AB and CD to be *congruent* if they have the same length. Here *length* is defined with the usual distance formulas in the first four cases, and the *length* of PQ in $\mathbb{S}$ is measured along the shortest $\mathbb{S}$-line from P to Q. We will study models of geometry in which the congruence axioms listed below are satisfied.

Segment Duplication Axiom. *Given segment AB and ray $\overrightarrow{CD}$, there is a unique point E on $\overrightarrow{CD}$ such that $AB \cong CE$.*

Segment Addition Axiom. *Suppose that $[ABC]$ and $[DEF]$. If $AB \cong DE$ and $BC \cong EF$, then $AC \cong DF$.*

Segment Equivalence Axiom. *For any segments AB, CD, and EF,*

(i) *$AB \cong AB$;*

(ii) *if $AB \cong CD$, then $CD \cong AB$;*

(iii) *if $AB \cong CD$ and $CD \cong EF$, then $AB \cong EF$.*

Problem 25. *Which of the segment congruence axioms hold in each of the models $\mathbb{R}^2$, $\mathbb{R}^3$, $\mathbb{Q}^2$, $\mathbb{D}$, and $\mathbb{S}$?*

Segment Subraction Proposition 26. *Assume that $[ABC]$ and $[DEF]$. If $AB \cong DE$ and $AC \cong DF$, then $BC \cong EF$.*

Segment Division Proposition 27. *Assume that* $[ABC]$ *and* $AC \cong DF$. *Then there is a unique point* E *between* D *and* F *such that* $AB \cong DE$ *and* $BC \cong EF$.

The congruence relation on segments gives us a natural way to order segments. We say that $AB < CD$ (or $CD > AB$) if there is a point E between C and D such that $AB \cong CE$.

Segment Ordering Proposition 28. *For points* A, B, C, D, E, F *with* $A \neq B$, $C \neq D$, *and* $E \neq F$,

(i) $AB \not< AB$;

(ii) *if* $AB > CD$ *and* $CD \cong EF$, *then* $AB > EF$;

(iii) *if* $AB < CD$ *and* $CD \cong EF$, *then* $AB < EF$;

(iv) *if* $AB < CD$ *and* $CD < EF$, *then* $AB < EF$;

(v) *exactly one of the following is true:* $AB < CD$, $AB \cong CD$, $AB > CD$.

Our last undefined notion is that of *congruence of angles*. We denote the angle which is the union of the non-opposite rays $\overrightarrow{AB}$ and $\overrightarrow{AC}$ by $\angle CAB$ or, if the context makes it unambiguous, by $\angle A$. We write

$$\angle CAB \cong \angle C'A'B'$$

to indicate that the angles are congruent. Combining segment and angle congruence leads to the defined notion of triangle congruence. Triangles $\triangle ABC$ and $\triangle A'B'C'$ are ***congruent***, written

$$\triangle ABC \cong \triangle A'B'C',$$

if under the correspondence $A \leftrightarrow A'$, $B \leftrightarrow B'$, and $C \leftrightarrow C'$ between their vertices, the angles $\angle A$, $\angle B$, and $\angle C$ are congruent, respectively, to the angles $\angle A'$, $\angle B'$, and $\angle C'$; and the sides AB, BC, and AC are congruent, respectively, to the sides $A'B'$, $B'C'$, and $A'C'$. Notice that the truth of the statement "$\triangle ABC \cong \triangle A'B'C'$" depends on the order in which the vertexes are listed!

In each of our models, we say that $\angle A$ and $\angle B$ are congruent if they have the same degree measure where, in $\mathbb{S}$, degrees are measured between the tangent lines. To our list of models, we now add a collection of new ones.

$\mathbb{B}$ (the Bumpy Planes)

Begin with the plane $\mathbb{R}^2$, and imagine heating it like a pizza in the oven, so that the surface forms an irregular pattern of bumps and valleys. Points,

lines, and betweenness are the same as in $\mathbb{R}^2$, but lengths and angle measures are changed. We say that two segments are congruent if they have the same length, and that two angles are congruent if they have the same degree measure. Note that there are many different Bumpy Planes because there are many different ways to arrange the bumps and valleys.

Every bumpy plane satisfies the betweenness axioms, since betweenness is exactly the same as in $\mathbb{R}^2$. It is easy to check that the segment congruence axioms are also satisfied in each $\mathbb{B}$.

Bumpy planes can be thought of as better models of irregular surfaces, like that of the earth, than $\mathbb{R}^2$. According to general relativity, space is not flat like $\mathbb{R}^3$, but rather is warped into bumps around massive objects. Though bumpy planes model physical reality better than does $\mathbb{R}^2$, this advantage comes with the price that they are generally much more complex than $\mathbb{R}^2$.

We now list our final axioms for how we like angle congruence to behave.

Angle Duplication Axiom. *Given angle $\angle CAB$ and ray $\overrightarrow{A'B'}$, there is a unique ray $\overrightarrow{A'C'}$ on a given side of line $\overleftrightarrow{A'B'}$ such that $\angle CAB \cong \angle C'A'B'$.*

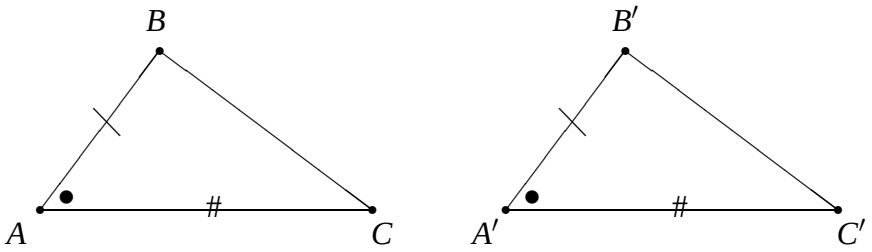
Angle Equivalence Axiom. *For any angles $\angle A$, $\angle B$, and $\angle C$,*

(i) *$\angle A \cong \angle A$;*

(ii) *if $\angle A \cong \angle B$, then $\angle B \cong \angle A$;*

(iii) *if $\angle A \cong \angle B$ and $\angle B \cong \angle C$, then $\angle A \cong \angle C$.*

SAS Congruence Axiom. *If two sides and the included angle of one triangle are congruent, respectively, to the corresponding two sides and included angle of another triangle, then the two triangles are congruent.*



Notice that the conclusion of SAS is, by definition, an abbreviation for six assertions, so SAS tells us that certain triples of those six imply the other three. In this sense, SAS is similar to the two Four Point Axioms.

Problem 29. *Which of the angle congruence axioms are satisfied in each of our models? Which models satisfy all of the incidence, betweenness, and congruence axioms?*

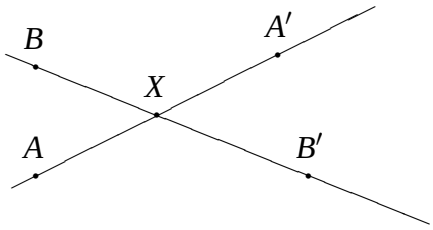
A triangle is said to be ***isosceles*** if two of its sides are congruent.

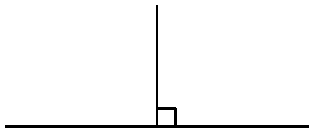
Proposition 30. *The angles opposite to the congruent sides of an isosceles triangle are congruent.*

A ***supplement*** of an angle $\angle CAB$ is an angle $\angle CAD$ where $[B,A,D]$.

Proposition 31. *Congruent angles have congruent supplements.*

If $[AXA']$ and $[BXB']$ but these points are not all collinear, then we say that the angles $\angle AXB$ and $\angle A'XB'$ are ***vertical angles***. An angle that is congruent to one of its supplements is called a ***right angle***.





Proposition 32. *Vertical angles are congruent to each other.*

Proposition 33. *An angle congruent to a right angle is a right angle.*

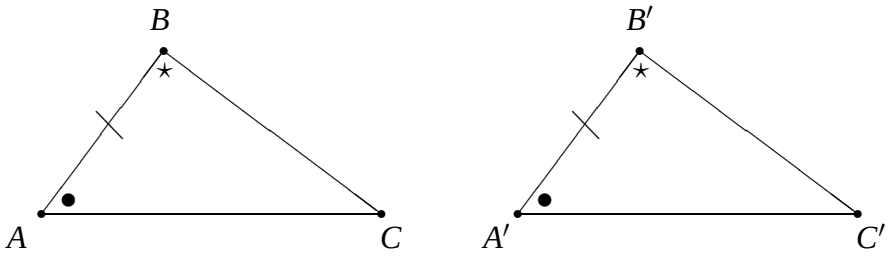
Proposition 34. *Assume that two lines ℓ and m intersect at a point P. If one of the four angles formed is a right angle, then so are the other three.*

Under the conditions of Proposition 34, we say that lines ℓ and m are ***perpendicular***.

Proposition 35. *Given a line ℓ and a point A, there is a line through A that is perpendicular to ℓ.*

Do you think the line described in Proposition 35 is unique?

ASA Congruence Theorem 36. *If two angles and the included side of one triangle are congruent respectively to the corresponding two angles and included side of another triangle, then the two triangles are congruent.*

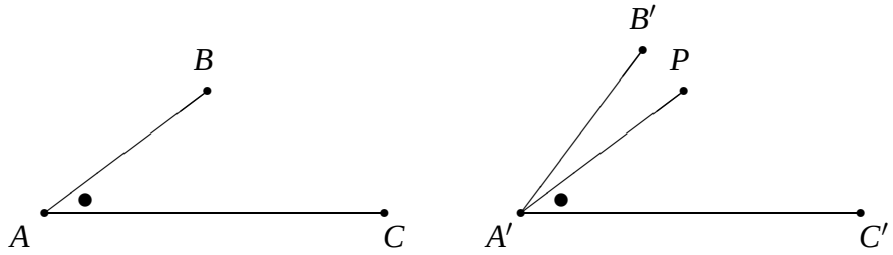


Proposition 37. *If two angles of a triangle are congruent, then so are the two opposite sides* (*that is, the triangle is isosceles*).

Angle Addition Proposition 38. *Given point D in the interior of $\angle CAB$, and point D' in the interior of $\angle C'A'B'$, if $\angle CAD \cong \angle C'A'D'$ and $\angle DAB \cong \angle D'A'B'$, then $\angle CAB \cong \angle C'A'B'$.*

Angle Subraction Proposition 39. *Given point D in the interior of $\angle CAB$, and point D' in the interior of $\angle C'A'B'$, if $\angle CAB \cong \angle C'A'B'$ and $\angle DAB \cong \angle D'A'B'$, then $\angle CAD \cong \angle C'A'D'$.*

Ordering of angles is defined just as we defined ordering of segments. We say that $\angle CAB < \angle C'A'B'$ if there is a point P in the interior of $\angle C'A'B'$ such that $\angle CAB \cong \angle PA'B'$.



Angle Ordering Proposition 40. *For angles $\angle A$, $\angle B$, $\angle C$,*

(i) *exactly one of the following is true: $\angle A < \angle B$, $\angle A \cong \angle B$, $\angle A > \angle B$;*

(ii) *if $\angle A < \angle B$ and $\angle B \cong \angle C$, then $\angle A < \angle C$;*

(iii) *if $\angle A > \angle B$ and $\angle B \cong \angle C$, then $\angle A > \angle C$;*

(iv) *if $\angle A < \angle B$ and $\angle B < \angle C$, then $\angle A < \angle C$.*

This proposition gives us a useful way to prove two angles are congruent: assume that one is larger than the other, and show that this leads to a contradiction. We look at two applications of this idea.

SSS Congruence Theorem 41. *If the sides of one triangle are congruent respectively to the sides of another triangle, then the two triangles are congruent.*

Proposition 42. *Any two right angles are congruent to each other.* (*Hint: Use Proposition 22*(*iii*).)

If lines *m* and *n* intersect a third line ℓ (called a ***transversal***) at two different points, then a pair of angles formed by *m* and ℓ and by *n* and ℓ that lie between *m* and *n* and on opposite sides of ℓ are called ***alternate interior angles***. (See Figure 4.1.)

Alternate Interior Angle Theorem 43. *If two lines intersect a transversal line in such a way that the alternate interior angles are congruent, then the two lines are parallel.*

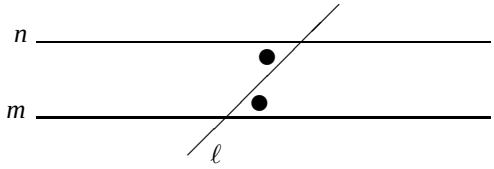


Figure 4.1: alternate interior angles

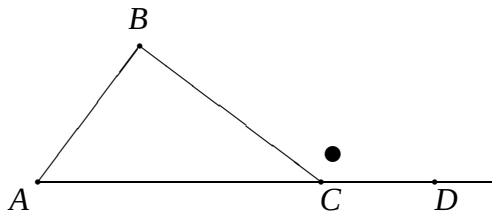
Notice that this theorem *does not* say the converse: if the lines are parallel, then the alternate interior angles must be congruent.

Proposition 44. *Two lines perpendicular to the same line are parallel. In particular, if A is not on ℓ, then the perpendicular to ℓ through A is unique.*

At the outset, we considered three possible ways in which parallel lines might behave uniformly, described by the Euclidean, Elliptic, and Hyperbolic Parallel Axioms. We can now show that our axioms prohibit the Elliptic Parallel Property.

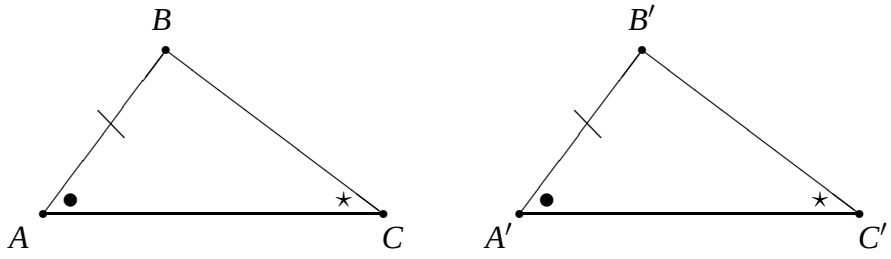
Parallel Line Theorem 45. *Given a line ℓ, and a point P not on ℓ, there is at least one line through P parallel to ℓ.*

A supplement of one of the three angles of a triangle is called an ***exterior angle*** of the triangle.



Exterior Angle Theorem 46. *An exterior angle of a triangle is greater than either remote interior angle.*

SAA Congruence Theorem 47. *If two angles and the side opposite one in a triangle are congruent, respectively, to two angles and the side opposite one in another triangle, then the two triangles are congruent.*



Problem 48. *Give a counterexample to show that SSA is not a valid criterion for congruence.*

Nevertheless, there is a special case when SSA is valid, namely, if the angle is a right angle. A ***right triangle*** is a triangle containing a right angle. The ***hypotenuse*** of a right triangle is the side opposite the right angle, and the other two sides are called the ***legs***.

HL Congruence Theorem 49. *Two right triangles are congruent if the hypotenuse and a leg of one are congruent to the hypotenuse and a leg of the other.*

We say that P is a ***midpoint*** of segment AB if $[APB]$ where $AP \cong PB$. We say that the ray $\overrightarrow{AP}$ is a ***bisector*** of $\angle CAB$ if P is in the interior of $\angle CAB$ where $\angle CAP \cong \angle PAB$.

Midpoint Theorem 50. *Every segment has a unique midpoint.*

Bisector Theorem 51. *Every angle has a unique bisector.*

This course has developed most of the fundamental facts of Euclidean geometry from Hilbert's axioms. Continuing in this same manner, with time and patience, it is possible to develop the entire subject.

Axioms

Symmetry Axiom. *If $[ABC]$, then A, B and C are distinct points and $[CBA]$.*

Uniqueness Axiom. *Given three points, at most one is between the other two.*

Collinearity Axiom. *If three points are collinear, then one is between the other two.*

1st 4-Point Axiom. *If $[ABC]$ and $[ACD]$, then $[ABCD]$.*

2nd 4-Point Axiom. *If $[ABC]$ and $[BCD]$, then $[ABCD]$.*

Infinity Axiom. *If B and D are distinct points, then there are distinct points A, C, and E on line $\overleftrightarrow{BD}$ such that $[ABCDE]$.*

Triangle Axiom. *There exists a triangle.*

Plane Separation Axiom.

(i) *If $A,B\,|_m$ and $B,C\,|_m$, then $A,C\,|_m$.*

(ii) *If $A\,|_m\,B$ and $B\,|_m\,C$, then $A,C\,|_m$.*

Segment Duplication Axiom. *Given segment AB and ray $\overrightarrow{CD}$, there is a unique point E on $\overrightarrow{CD}$ such that $AB \cong CE$.*

Segment Addition Axiom. *Suppose that $[ABC]$ and $[DEF]$. If $AB \cong DE$ and $BC \cong EF$, then $AC \cong DF$.*

Segment Equivalence Axiom. *For any segments AB, CD, and EF,*

(i) *$AB \cong AB$;*

(ii) *if $AB \cong CD$, then $CD \cong AB$;*

(iii) *if $AB \cong CD$ and $CD \cong EF$, then $AB \cong EF$.*

Angle Duplication Axiom. *Given angle $\angle CAB$ and ray $\overrightarrow{A'B'}$, there is a unique ray $\overrightarrow{A'C'}$ on a given side of line $\overleftrightarrow{A'B'}$ such that $\angle CAB \cong \angle C'A'B'$.*

Angle Equivalence Axiom. *For any angles $\angle A$, $\angle B$, and $\angle C$,*

(i) *$\angle A \cong \angle A$;*

(ii) *if $\angle A \cong \angle B$, then $\angle B \cong \angle A$;*

(iii) *if $\angle A \cong \angle B$ and $\angle B \cong \angle C$, then $\angle A \cong \angle C$.*

SAS Congruence Axiom. *If two sides and the included angle of one triangle are congruent, respectively, to the corresponding two sides and included angle of another triangle, then the two triangles are congruent.*

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